Minimal Cover : (Canonical Cover)

It's a simplified & reduced version of the given set of FDs. Since, it is a reduced version, it is also called as irreducible set. It is also called as canonical cover.

Steps to find Min. Cover :

1. Split the Right hand attributes of all FDs to single value.
2. Remove the extraneous attributes.
3. Remove the redundant FDs.

Ex: $AB \rightarrow C$, either A or B can be extraneous.

If A closure contains B, then B is extraneous and can be removed.

2012

He who has a 'why' to live, can bear with almost any 'how'.

You can

Minimal Cover

Q: $A \to BC$, $B \to C$, $A \to B$, $AB \to C$ — $R(ABC)$, $F_c = ??$

1) Single attribute

$A \to B$
$A \to C$
$B \to C$
$A \to B$
$AB \to C$

II) Remove redundant attributes

$AB \to C$

Remove A, $B^+ = BC$
Remove B, $A^+ = ABC$

$A \to B$
$A \to C$ ✗
$B \to C$
$A \to B$
$A \to C$ ✗

III) Remove redundant FDs.

$A \to B$ & $B \to C \Rightarrow A \to C$

Remove $B \to C$, $B^+ = ??$
Remove $A \to B$, $A^+ = ??$

Hence, $F_c = \{B \to C, A \to B\}$

FEB 2012

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29			

Key Problem

Q- $R(ABCDE)$

$F = \{A \to BCD, B \to E, AC \to BDE, D \to A\}$

Find the types of keys.

A- Finding the closure of all LHS of FDs.

$A^+ = ABCDE$
$B^+ = BE$
$AC^+ = ACBDE$
$D^+ = ABCDE$

Keys → A, AC, D

Candidate key → A, D
Super key → AC (as C is an extraneous attribute / additional attribute)

Q $R(ABCDEH)$

$F = \{A \to BC, CD \to E, E \to C, D \to AEH, ABH \to BD, DH \to BC\}$

You can only find truth with logic if you have already found truth without it.

A. Finding the closure of all LHS of FDs,

$A^+ = ABC$
$CD^+ = CDEAHB$
$E^+ = EC$
$ABH^+ = ABHEDC$
$DH^+ = DHBCEA$
$D^+ = DAEHBC$

Keys → CD, D, ABH, DH

Candidate Key → D
Super Key → CD, DH

Now, testing whether ABH is a super key or candidate key,

Find AB^+, AH^+, BH^+

$AB^+ = ABC$
$AH^+ = AHBCDE$
$BH^+ = BH$

⇒ As, AH is found to be a candidate key, ABH is a super key with B as an extraneous attribute.

Hence,

S. K → ABH, CD, DH
C. K → D, AH

S M T W T F S
1 2 3 4
5 6 7 8 9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29

Algorithm to Find a key K for R given a set of FDs F

1. let K = R
2. For each attribute A in K
{
Compute $(K-A)^+$ w.r.t F
if $(K-A)^+$ contains all the attributes in R,
then, let K : K−{A}
}

Q. R(ABCD)

$F = \{A \to C, A \to B, BC \to D\}$

Find the key through above algorithm.

A: let, $K = ABCD = R$.

i) $(K-A)^+ = (BCD)^+ = BCD \neq R$.

⇒ K remains as previous i.e K = ABCD

ii) $(K-B)^+ = (ACD)^+ = ACDB = R$.

⇒ k is updated as

$K = K - B = ACD$.

Human history becomes more and more a race between education and catastrophe.

iii) $(K-C)^+ = (AD)^+ = ADBC = R$

$\Rightarrow$ K is updated as,

$$K = K - C = AD$$

iv) $(K-D)^+ = (AD-D)^+ = A^+ = ABCD = R$

$\Rightarrow$ K is finally updated as,

$$K = K - D = A$$

Hence, A is the candidate key derived.

Non-Redundant Cover

→ It finds the duplicated FDs.

Algorithm :

Input → A set of FDs, F.
Output → A non-redundant cover of F.

$G := F$
for each FD, $X \to Y$ in G do
    if $X \to Y \in \{F - (X \to Y)\}^+$
    then, $\{F = F - (X \to Y)\}$
$G := F$

Q. $R(ABCDEH)$

$F = \{ A \to BC$
$CD \to E$
$E \to C$
$D \to AEH$
$ABH \to BD$
$DH \to BC \}$

Step 1: Test for $\boxed{A \to BC}$,

Find A^+ under $\{F - (A \to BC)\}$

$A^+ = A$

Since $BC \not\subseteq A^+$, so, $A \to BC$ is non-redundant and cannot be removed.

Step 2: Test for $\boxed{CD \to E}$,

Find CD^+ under $\{F - (CD \to E)\}$

$CD^+ = CDAEHB$

Since, $E \subseteq CD^+$, so, $CD \to E$ is redundant and will be removed

Now, $F = \{ A \to BC, E \to C, D \to AEH,$
$ABH \to BD, DH \to BC \}$

The secret of success is to know something nobody else knows.

Step 3 : Test for $E \to C$,

Find E^+ under $\{F - (E \to C)\}$

$E^+ = E$

Since, $C \notin E^+$, so, $E \to C$ is non redundant & cannot be removed.

Step 4 : Test for $D \to AEH$,

Find D^+ under $\{F - (D \to AEH)\}$

$D^+ = D$

Since, $AEH \notin D^+$ so, $D \to AEH$ is non-redundant & cannot be removed.

Step 5 : Test for $ABH \to BD$,

Find ABH^+ under $\{F - (ABH \to BD)\}$

$ABH^+ = ABHC$

Since, $BD \notin ABH^+$, so, $ABH \to BD$ is non-redundant & cannot be removed.

Step 6 : Test for $DH \to BC$,

S M T W T F S
1 2 3
4 5 6 7 8 9 10
11 12 13 14 15 16 17
18 19 20 21 22 23 24
25 26 27 28 29 30 31

Find DH^+ under $\{F - (DH \rightarrow BC)\}$

$DH^+ = DHAEBC$

Since $BC \subseteq DH^+$, so, $DH \rightarrow BC$ is redundant & will be removed.

Hence, the Non-Redundant cover of F

$= \{A \rightarrow BC, E \rightarrow C, D \rightarrow AEH, ABH \rightarrow BD\}$

Practice Questions

1) Minimize $\{A \rightarrow C, AC \rightarrow D, E \rightarrow H, E \rightarrow AD\}$

2) Find the canonical cover F_c of the given set, $F = \{AB \rightarrow C, D \rightarrow E, AB \rightarrow E, E \rightarrow C\}$.

3) Minimize $\{A \rightarrow C, AB \rightarrow C, C \rightarrow DI, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C\}$

4) Find the non-redundant cover of the following set of FDs defined on the relation R(BCDEFG).
$F = \{CF \rightarrow BG, G \rightarrow D, CDG \rightarrow F, B \rightarrow DE, D \rightarrow C\}$

The artist is nothing without the gift, but the gift is nothing without work.